Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/03/2025
PI Name:	Robertson, Ian Ph.D.		
Project Title:	Human-System Performance Verification		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/11/2024	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per the D. Kulkarni/HFBP (Ed., 6/5/25).		
Key Personnel Changes/Previous PI:	Note: Per the Principal Investigator (PI), Annlyle Diokno, Ph.D. was added to the investigation to conduct data collection with participants and assist with execution of the experimental procedure, study documentation, methodology design, and analysis of error-related data (Ed., 10/23/25).		
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Task Description:	Spaceflight programs verify vehicle readiness for flight based on compliance with hundreds of requirements. For some requirements (such as those governing usability, workload, and errors), individual differences matter, and a representative sample of the astronaut population must be tested in a scenario-based Human-in-the-Loop (HITL) evaluation. To date, the required number of test subjects for these HITLs has been determined by NASA subject matter experts (SMEs), based on available literature. If design issues are encountered in flight, particularly during an emergency or stressful activity, the issues could result in errors and have safety impacts. In addition, the fewer subjects tested, the greater the variability in the data, and the lower the confidence in the results. The debate over how many subjects is sufficient for a test is neither new, nor is it limited to NASA HITLs. A paper frequently cited by programs and providers states that 5 subjects are sufficient for discovering usability problems (Nielsen, 2000). But this recommendation is qualified on the assumption that the tests are of a qualitative nature, and that they are part of a series of iterative tests, so that when taken together, the decision is based on more than just a sample of 5. Contrast this with a verification HITL, which is typically a single test event with quantitative outcomes. Nielsen (2012) is also on the record stating that a minimum of 20 participants should be involved in tests using quantitative outcomes. The purpose of the proposed task is to reduce future program risks by providing high-fidelity spaceflight HITL dataset and analysis examples with varying sample sizes to better inform NASA verification decisions. The goal is to demonstrate how verification test results and threshold decisions may change as sample size goes from two to 20. It is anticipated that these data will be more relevant and informative to NASA management than the data presented from the literature to date.
Rationale for HRP Directed Research:	This proposal qualifies for directed study due to the insufficient time required for solicitation and the highly constrained nature of the work. The proposers are currently involved to varying degrees in all of the commercial programs as subject matter experts and are well aware of the commercial arguments that have been made, and the Human Health and Performance evidence that has been presented to management from the literature. Note that much of the research data on recommended Human-in-the-Loop (HITL) sample size comes from the domain of consumer products and website design, rather than safety-critical environments. The proposers are aware of the commercial designs, design issues, and test plan approaches, and have technical relationships with some of these teams that will facilitate the project. External researchers would not have the specific background knowledge, nor access to commercial vehicle information and technical contacts; thus, directed research would be more cost effective, and the results would have greater validity.
Research Impact/Earth Benefits:	This project stands to inform future discussions about how sample size impacts the validity of test results.
Task Progress:	Human-in-the-loop (HITL) tests play a critical role in the development and certification of space systems. An important aspect of planning a HITL test is determining the appropriate sample size needed for the test. While NASA requirements often specify the number of subjects required for verification, recently there has been an increase in requests to use fewer subjects to certify spacecraft. The NASA Human Health and Performance subject matter expert (SME) community has serious concerns about this trend. Exacerbating the issue is that many rules-of-thumb available to guide sample size selection exist within the literature. However, it's possible that these rules are cited out of context, misunderstood, and/or misapplied. Further, there is a lack of data relevant to spaceflight to inform discussions regarding sample size for HITL verifications. The goal of this project is to inform current and future discussions on the appropriate sample size for performing HITL tests at NASA. The aims of this project were: 1. Sample Size Rules-of-Thumb: Understand what sample size rules-of-thumb are available in the literature, and how they may or may not apply to spaceflight HITL verification tests. 2. Sample Size Lessons Learned: Perform knowledge elicitation with relevant internal and external experts on sample size for the verification and summarize them as lessons learned for NASA spaceflight programs. 3. Impact of Sample Size on Verification Outcomes: Investigate how sample size impacts data characteristics for selected verification outcomes, systems, and tasks. Aim 1: Sample Size Rules-of-Thumb To fulfill Aim 1, we identified 31 relevant articles containing sample size rules-of-thumb. These articles were summarized in terms of their context of use, the population used to generate the data, and their recommended application. The results of the synthesized literature can be summarized as follows. Sample size rules-of-thumb vary in context of use and recommended application. Many of the rules-of-thumb available in the literature are not applicable to HITL verification tests due to specific context of use or because they are aimed at outcomes not used in HITL verification tests. Finally, it's clear from the available literature that there exists a gap for sample-size-guidelines pertinent to aerospace HITL testing. Aim 2: Lessons Learned To complete Aim 2, we conducted interviews with representatives from different stakeholders' groups. The objectives were to identify factors that impacted past decisions, leading to smaller sample sizes, and recommendations for future decisions related to how many subjects to use for HITL verification tests. From these interviews, we identified several contextual factors that may lead to the selection of smaller sample sizes. Aim 3: Impact of Sample Size on Verification Outcomes To fulfill Aim 3, we collected data via an experimental study using two scenarios. In both scenarios, participants completed a verification-like scenario using a high-fidelity spaceflight mockup. The study consisted of two sessions. In the first session, participants were familiarized with the scenario, task procedure, and interface. They then completed a practice run. In the second session, participants reviewed the learning material and then they completed a final run-for-record. Participants' primary task was to execute either a manual or automated procedure. Participants also completed a secondary task in the form of monitoring flight subsystems and/or alarm systems.

	Based on the results of our analyses we conducted, we present the following general conclusions: •The risk of making an incorrect decision is higher for smaller sample sizes o This was particularly true for sample sizes of N = 3 and N = 5 •The risk of an incorrect decision is small when the population values are far away from the decision criteria threshold o The risk of making an incorrect decision is much higher when the population value is closer to the decision criterion threshold •The NASA Modified System Usability Scale was particularly vulnerable to incorrect decisions at smaller sample sizes o Probably a function of average vs. highest observed value pass criteria •Bedford Workload and Error Rates were more resistant to false pass/fail decisions, though they still occurred o Both criteria are more conservative by requiring all observed values to pass the threshold •More data will generally lead to a better decision o No magic number for preventing, but N = 10 and N = 15 were inflection points many times and may provide a starting point for future sample size discussions at NASA
Bibliography Type:	Description: (Last Updated:)